

Mandromy data of semi-simple Frobenius manifolds

①

→ local invariant of the Frobenius manifold.

domain $\mathcal{SCH}(X)$

Why do we care?

Dubrovin's conjecture: $X \xrightarrow{K_X \text{ simple}} \text{Fano variety}$, with $F(t)$ convergent in some

- $QH(X)$ semi-simple $\iff \exists$ full exceptional collection of $D^b(X)$
- Stokes matrix $S \iff$ Euler pairing $\chi(E_i, E_j)$
- Central connection matrix $C \iff \sum_i (-1)^k \dim_c \text{Hom}(E_i, E_j[k])$
Chern characters $Ch(E_i)$.

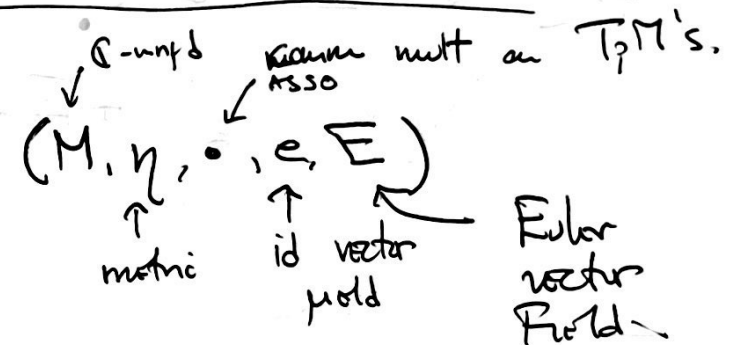
Recap of basics

Recall that a Frobenius mfd is a tuple such that: $(T_p M, \star_p, \eta_p)$ Frobenius alg with unit e_p .

- η is flat ($F_{\nabla} \eta = 0$).

- If $c(x, y, z) = \eta(x \star y, z)$ then $(\nabla_w c)(x, y, z)$ is sym in all x, y, z, w .

- $\nabla e = 0$,



• $\mathcal{L}_E \eta = (2-D)\eta$, $D \in \mathbb{C}$, $\mathcal{L}_E(*) = *$ and $\nabla(\nabla E) = 0$. (2)

Last time, we introduced the connection on M

$$\tilde{\nabla}_X Y = \nabla_X Y + Z X * Y \quad Z \in \mathbb{C}$$

↖ L.C. connection

For a Frob. mfd $\tilde{\nabla}$ is flat and torsion free.
 $\Rightarrow$ If t^α are flat coords per ∇ , $\exists \tilde{t}^\alpha$ flat coords per $\tilde{\nabla}$
 and $\tilde{t}^\alpha = (t, Z)$.

If $\pi: M \times \mathbb{C} \rightarrow M$, we can extend $\tilde{\nabla}$ to flat conn. on
 $\pi^*(TM) \rightarrow M \times \mathbb{C}$ by defining

$$\tilde{\nabla}_{\partial_Z} Y = \partial_Z Y + E * Y - \frac{1}{Z} \left(\frac{2-D}{2} Y - \nabla_Y E \right)$$

$V(Y)$

Prop: if starting from $(M, \eta, * \leftarrow e, F)$ vector field such that

- $\tilde{\nabla}$ is flat
 - $\mathcal{L}_E \eta = (2-D)\eta$
 - $\nabla e = 0$
- then (M, η) is Frobenius.

Two perspectives

- Fix $z \in \mathbb{C}$, $\tilde{\nabla}$ flat torsion free connection on $TM \rightarrow M$
- Fix $p \in M$, $\tilde{\nabla}_z$ flat monomorphic connection on $P' \times T_p M \rightarrow P'$

$$\tilde{\nabla}_z Y = \partial_z Y + E * Y - \frac{1}{z} V(Y) \quad \text{regular sing at } z=0.$$

$$\text{with } \omega = \frac{1}{z} \quad \tilde{\nabla}_\omega Y = \partial_\omega Y - \frac{1}{\omega^2} E * Y + \frac{1}{\omega} V(Y) \quad \text{reg sing at } z=\infty.$$

Monodromy data at $z=0$.

We focus on solutions of $\tilde{\nabla}_{\partial_z} \xi^\beta = 0$ near $z=0$. $\beta = 1, 2, \dots, \dim_{\mathbb{C}} M$, $\tilde{\nabla}_{\partial_z} d\tilde{t}^\beta$

~~After identifying $TM \cong T^*M$ via η we fix ~~frame~~ flat~~
 $\tilde{\nabla}_{\partial_z} \xi = 0$ becomes $\partial_z \xi =$ coords t^α mod p .

diagonalizing ∇E , $\nabla E = (1 - q^\alpha) \partial_{t^\alpha}$

$$\text{We denote } \nu = \text{diag} \left(\frac{2-D}{2} - (1-q^1), \dots, \frac{2-D}{2} - (1-q^n) \right) \\ = \text{diag} \left(q^1 - D/2, \dots, q^n - D/2 \right)$$

After identifying $TM \rightarrow T^*M$ via η , the equation (4)

$\tilde{\nabla}_{\partial_z} dt^\alpha = 0$ becomes $\partial_z \xi^\alpha = E^\epsilon C^\alpha_{\epsilon\beta} \xi^\beta - \frac{1}{z} \nu^\beta \xi^\beta$.

$\xi = \xi^\beta \partial_{t^\beta}$.

$\partial_z \xi = \left(\nu + \frac{1}{z} \mu \right) (\xi)$. $\mu = \text{diag} \left(q^\alpha - \frac{2}{D} \right)$
 $\partial_z \xi^\alpha = E^\epsilon C^\alpha_{\epsilon\beta} \xi^\beta + \frac{1}{z} \mu^\alpha_\beta \xi^\beta$ μ^α_β

ν is η -symmetric, while μ is η -antisymmetric:

Prop: there is a gauge transformation $G(z) = 1 + z G_1 + \dots$

$G^T(-z) \eta G(z) = \eta$.

such that for $\tilde{\xi} = G(z) \xi$

$\partial_z \tilde{\xi} = \left(\frac{1}{z} \mu + R_1 + R_2 z + \dots \right) \tilde{\xi}$ (*)

and $\dots R_k$ is η -sym (resp η -anti-sym) if k -odd (resp. k -even)

$(R_k)^\alpha_\beta \neq 0$ only if $\mu^\alpha - \mu_\beta = k$.

For (*) we have an easy fundamental sol. given by $z^\nu z^R$ $R = R_1 + R_2 + \dots$

Indeed $\partial_z (z^\nu z^R) = \frac{\nu}{z} z^\nu z^R + z^\nu \frac{R}{z} z^R = \left(\frac{\nu}{z} + \frac{z^\nu R z^{-\nu}}{z} \right) z^\nu z^R$

but note $(z^N R_K z^{-N})^\alpha_\beta = z^{N\alpha - \mu\beta} R_K^\alpha_\beta = z^K (R_K)^\alpha_\beta$ (5)

$\Rightarrow z^N R_K z^{-N} = z^K R_K \Rightarrow \frac{z^N R z^{-N}}{z} = (R_1 + z R_2 + \dots)$

Rmk: the form (*) is not uniquely determined

Here a solution to $\partial_z \xi = (V + \frac{1}{z} \mu) \xi$ has the form

$$\xi = G(z)^{-1} z^\mu z^R.$$

It can be shown that R commutes with $\exp(2\pi i \mu)$,
so monodromy of ξ around $z=0$ is $\exp(2\pi i (\mu + R))$.

~~then Def: the monodromy data is covariantly constant in p .~~

Def: the monodromy data ~~at $z=0$~~ ^{near $z=0$} of ~~(R, η, μ)~~ at $(p, 0) \in M$ of $(M, \eta, \mu, e, \nabla)$ is $(T_p M, \eta_p, \mu_p, [R]_p)$.

then, the data $(T_p M, \eta_p, \mu_p, [R]_p)$ ~~does not depend on $p \in M$.~~
is covariantly constant in p .

Then: we can find plot coords $\tilde{t}^\alpha$ per M round p.c.h. (6)
 such that for z suff small

$$\tilde{t}^\alpha(t, z) = h^\alpha(t, z) z^p z^R$$

$$h^\alpha(t, z) = \cancel{\sum_{k=0}^p} t^\alpha + \sum_{k=1}^{\infty} h_k^\alpha(t) z^k.$$

Monodromy Around $z = \infty$.

We consider again the system $\partial_z \xi = z C_\alpha \xi$ (a)
 $\partial_z \xi = \left(\nu + \frac{1}{z} \rho \right) \xi$

we fix a point $p \in M$ such that $y \rightarrow E \cdot y$ has pairwise
 distinct eigenvalues v_i

Around p , we have canonical coords v_i s.t

$$\frac{\partial}{\partial v_i} \cdot \frac{\partial}{\partial v_j} = \delta_{ij} \frac{\partial}{\partial v_i}$$

we consider the frame of normalized ~~element vectors~~ admissible

$$f_i = \frac{\partial v_i}{\|\partial v_i\|_g}$$

In the frame $(f_1, f_2, \dots, f_n)$

(2) becomes

(7)

$$(2) \quad \partial_{\bar{z}} \gamma = \left(U + \frac{V}{\bar{z}} \right) \gamma$$

$$\partial_{z'} \gamma = (z E_i + V_i) \gamma$$

where $U = \text{diag}(u_1, \dots, u_n)$

$$V^T + V = 0$$

$$(E_i)^a_b = \delta_i^a \delta_{ib}$$

V_i uniquely determined by

$$\gamma = \psi \xi \quad \psi$$

$$[U, V_i] = [E_i, V]$$

$$U = \psi \eta \psi^{-1}$$

$$V = \psi \mu \psi^{-1}$$

Prop There is a ~~unique~~ formal

gauge transformation series $G(z)$

$$G(z) = 1 + \frac{A_1}{z} + \frac{A_2}{z^2} + \dots$$

$$G^+(-z) G(z) \quad (3)$$

such that ~~(2)~~ (2) becomes

$$\partial_z \tilde{\gamma} = U \tilde{\gamma} \quad \tilde{\gamma} = G(z) \gamma$$

A solution to (3) is given by $\tilde{\gamma} = e^{zU}$

In the frame of idempotents a formal fundamental solution is then given by $\gamma_{\text{formal}} = G(z)^{-1} e^{zU}$

the Stokes rays associated to (2) are

(8)

$$R_{ij} = \{z \in \mathbb{C} / z = -ir(\bar{u}_i - \bar{u}_j), r \geq 0\}.$$

for $z \in R_{ij}$ $|e^{zu_i}| = |e^{zu_j}|$, for $z \in R_{ij}$ to the right of R_{ij} $|e^{zu_i}| < |e^{zu_j}|$, for z to the left $|e^{zu_i}| > |e^{zu_j}|$.

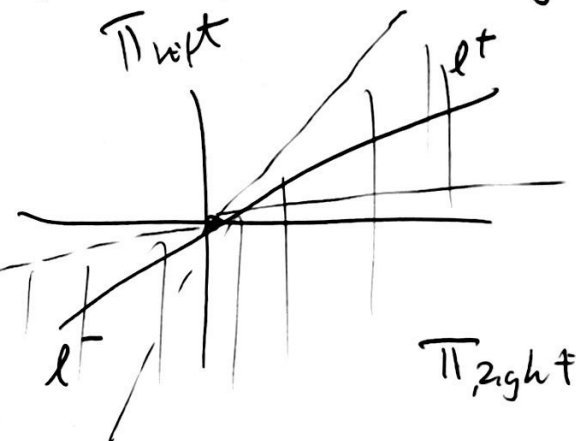
$$(|e^{zu_i}| < |e^{zu_j}| \Rightarrow 1 < |e^{z(u_j - u_i)}| \Rightarrow \operatorname{Re}(z(u_j - u_i)) > 0.$$

$$z = -ir(\bar{u}_i - \bar{u}_j)e^{i\theta} \quad \operatorname{Re}(+ir|u_j - u_i|^2 e^{i\theta}) > 0 \\ -\sin \theta r |u_j - u_i|^2 > 0$$

An admissible line l is a line through $0 \in \mathbb{C}$ not intersecting R_{ij} . Orient l , $l^+ = \{z \in l / \arg z = \phi\}$
 $l^- = \{z \in l / \arg z = \phi - \pi\}$.

and let $\pi_{\text{right}}, \pi_{\text{left}}$ the sectors $\rightarrow$

$$\pi_r \cap \pi_l = \bigcup_{l^+} \pi_+ \cup \bigcup_{l^-} \pi_-$$



Then there exists unique fundamental solns $Y_R(z), Y_L(z)$ of (2) such that $\textcircled{P}$
 $\underbrace{Y_{R/L}(z) \sim Y_{\text{formal}}(z)}_{\text{as } |z| \rightarrow \infty \text{ in } \mathbb{C} \setminus L}$

This means: for each n , $z^n \left| Y_{R/L}(z) - \sum_{k=0}^n \frac{A_k z}{z^k} \right| \rightarrow 0$

As $|z| \rightarrow \infty$ in $\mathbb{C} \setminus L$

On $\mathbb{C} \setminus L$ we have $Y_L(z) = Y_R(z) S_{\pm}$ for $S_{\pm}$

A constant ~~non~~ invertible matrix in $\mathbb{C}$

Remark: one has the relation $S_- = S_+^T$, so we usually just write $S := S_+$.

$S_{\pm}$ are called the Stokes matrices of (2).

Def: let $Y_0(z) = H(z) z^P z^R$ be a fundamental sol to (2) near $z=0$, with branch cut at l_-

the central connection matrix C is depends by

(10)

$$Y_0(z) = Y_{\text{right}}(z) C \quad z \in \Pi_+ \cap D_0$$

μ, R, S, C satisfy the compatibility relation.

$$C^{-1} S^T S^{-1} C = \text{Exp}(2\pi i (\mu + R))$$

then: (Isomonodromy part 2): S_p and C_p do not depend of p as long as no Stokes ray crossed ~~on~~ the admissible curve L .

the solutions $Y_{R,L}(z, p)$ depend holomorphically on u

and $\partial_{u_i} Y_{R,L} = (z E_i + \sum V_i) Y_{R,L}$ so they depend

flat ~~phases~~ ^{phases} of $\widehat{D}$.

$$\Rightarrow \partial_{u_i} S = 0 \quad \partial_{u_i} C = 0$$

Def: the monodromy ^{data} of $(M, \gamma, *, e, E)$ near $p \in M$ is $(T_p M, \gamma_p, \mu, [R])$ and S, C .

Example: $\mathbb{Q}H^*(\mathbb{P}^2)$

(11)

In this case, everything can be done explicitly.

We have global flat words $\phi = t^1 + t^2\omega + t^3\omega^2 \in H^*(\mathbb{P}^2)$

$\eta = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$, $\star$ is determined by

$$F(t) = \frac{1}{2} (t^1)^2 t^3 + \frac{1}{2} t^1 (t^2)^2 + \sum_{k=1}^{\infty} \frac{N_k}{(3k-1)!} (t^3)^{3k-1} e^{kt^2}$$

~~$N = \#$~~ $e = 1 \in H^0(\mathbb{P}^2)$

$$E = ((1-q^1)t^1 + r^1) \frac{\partial}{\partial t^1}$$

where ~~$q^1 = 1$~~ ~~$r^1 = 3$~~

$$q^0 = 0, q^1 = 1, q^2 = 2$$

$$\mathbb{D} = \text{div}_{\mathbb{P}^2} = 2$$

$N_k = \# \{ \text{rational curves of deg } k \text{ on } \mathbb{P}^2 \text{ passing through } 3k-1 \text{ generic points} \}$

And ~~$r^0 = 0$~~ $r^0 = r^2 = 0$ $r^1 = 3$

N and R are determined by purely classical DATA.

$$\nabla E = (1-q^1) \frac{\partial}{\partial t^1} \otimes dt^1$$

$$\mu^k = q^k - \frac{\mathbb{D}}{2} \quad \mu = \overset{\text{div}}{(-1, 0, 1)}.$$

In quantum cohomology, $R_i = 0$ for classical point $i > 1$ and R_1 is given by mult by $C_1(X)$ (12)

In our ~~case~~ ^{classical} case $C_1(P^2) = 3w$ so $R = R_1 = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$

We complete the Stokes matrix S at semi-simple point

$t^1 = t^3 = 0$ and t^2 is region of convergence of F

In frame $(\partial_1, \partial_2, \partial_3)$ E^* is given by $\begin{pmatrix} 0 & 0 & q \\ 3 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$ $q = e^{t^2}$

The canonical coords are given by $U_1 = 3q^{1/3}$, $U_2 = 3\bar{\epsilon}^2 q^{1/3}$
 $U_3 = 3\epsilon^2 q^{1/3}$
 $\epsilon = \exp(\frac{\pi i}{3})$

The transition function between $(\partial_1, \partial_2, \partial_3)$ and frame of normalized idempotents (f_1, f_2, f_3) is given by

$$\psi = \frac{1}{\sqrt{3}} \begin{pmatrix} q^{-1/3} & 1 & q^{1/3} \\ \bar{\epsilon} q^{-1/3} & -1 & \epsilon q^{1/3} \\ \epsilon q^{1/3} & -1 & \bar{\epsilon} q^{1/3} \end{pmatrix} \quad \gamma = \psi \epsilon$$

$\gamma^i f_i \quad S^{\alpha} \partial_{t^{\alpha}}$